

Project Overview

Project Title

****Readora – AI-Powered Digital Library System****

Objective

The objective of this project is to build a smart and user-friendly digital library platform that simplifies book searching, reservation, and tracking while providing intelligent assistance through an AI chatbot. The system improves organization, reduces manual workload, and enhances user experience for readers and library administrators.

Target Audience

- Students and academic readers
- Educational institutions and libraries
- Book lovers and cozy readers
- Small libraries seeking digital management solutions

Project Type

- AI-Augmented Learning & Education Platform
- Productivity System
- Small Business & Educational Support System

Tech Stack

Backend

- **Laravel (PHP Framework)**

Frontend

- **React.js** (Component-based UI development)

UI Design Tool

- **Canva** (Mock UI and Prototyping)

AI Integration

- **OpenAI API / Chatbot API (planned)**
- Used for intelligent chatbot assistance.

Database

- MySQL

Rendering Method

- **Client-Side Rendering (CSR)** using React.

UI Mock Design

The complete UI mockup has been designed in Canva.

Design Link:

https://www.canva.com/design/DAG95vQGKYI/Q5wdAvEY_bC2Ay8EZqPXoA/edit?utm_content=DAG95vQGKYI&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

Project Features

Authentication

- User Registration & Login
- JWT-based authentication (planned)

Book Management

- View all books
- Search books by title, author, category
- View availability status
- Reserve books

Intelligent Chatbot Assistant (Exclusive Feature)

- AI-powered chatbot helps users:
 - Find books instantly
 - Answer library related queries
 - Guide users through system navigation
 - Provide recommendations (future scope)
- Improves decision support and usability.

MyLibrary Dashboard

- View issued books
- Track due dates
- Monitor borrowing history

Category Browsing

- Organized book filtering by categories

CRUD Operations

Entity Operations
----- -----
Users Create, Read, Update
Books Create, Read, Update, Delete
Reservations Create, Read, Delete
Categories Create, Read
Issued Books Read, Update

API Endpoints (Approximate)

POST /api/auth/register

POST /api/auth/login

GET /api/books

GET /api/books/{id}

POST /api/books

PUT /api/books/{id}

DELETE /api/books/{id}

GET /api/categories

POST /api/reserve

GET /api/my-library

POST /api/chatbot/query

Milestones

Milestone 1 – UI & Planning

- Finalize UI mockups
- Setup GitHub repository
- Define database schema
- API planning

Milestone 2 – Core Development

- Laravel backend setup
- User authentication
- CRUD operations for books & categories
- Book reservation system

Milestone 3 – AI & Finalization

- Chatbot API integration
- Dashboard implementation
- Testing and bug fixing
- Final documentation

Bonus Features (Planned)

- AI chatbot integration
- Smart recommendations
- Notification alerts
- UI animation improvements
